Lecturers: Thomas Quella and Nick Beaton
School of Mathematics and Statistics

Second Semester, 2026
University of Melbourne

MAST20030: Differential Equations
Tutorial: Week 2

Question 1

(a) Find the general solution of the following ODE using an integrating factor:

$$y' + \frac{1}{t}y = \mathrm{e}^t. \tag{1}$$

(b) Using your understanding of the connection between linear algebra and differential equations, answer the following questions:

i. write down the linear operator, T, that is applied to y in (1).

ii. List both the complimentary solutions (i.e., the solution space of T) and particular solution you found when solving (1).

iii. What is the dimension of the solution space of T?

Question 2

In this question we are going to demonstrate how the process of reduction of order works by solving the ODE

$$\frac{1}{2}y'' + \sqrt{2}y' + y = 0.$$

(a) By trialling a solution of the form $y(t) = \mathrm{e}^{\lambda t}$, find a single solution to the ODE.

(b) By substituting a solution of the form $y(t) = u(t)\mathrm{e}^{\lambda t}$ into the ODE, where λ is specified from (a), find the general solution to the ODE. Do not use the reduction of order formula from the lecture notes but instead derive from first principles.

(c) Using the Wronskian, verify that your general solution contains the correct number of linearly independent solutions.

Question 3

By trialling $y = x^n$, find a solution of

$$x^2y'' - x(x+2)y' + (x+2)y = 0.$$

Use reduction of order to find the general solution to the ODE.

Question 4

In order to solve

$$xy'' - (1+x)y' + y = x^2\mathrm{e}^{2x},$$

show that $y = \mathrm{e}^x$ is a solution of the corresponding homogeneous equation and then apply reduction of order to find the general solution (of the inhomogeneous equation).